

Integration Techniques

Problem 1

Consider a differentiable function f with the following table of values.

x	0	1	$\pi/2$	e	3
$f(x)$	5	7	8	20	22
$f'(x)$	2	4	6	9	12

(a) Evaluate $\int_1^3 f'(x)e^{f(x)}dx$

(b) Evaluate $\int_0^1 e^x f'(e^x)dx$

(c) Evaluate $\int_0^1 f'(x)(f(x))^2dx$

Solution.

(a) We use the change of variables $u = f(x)$,

$$\frac{du}{dx} = f'(x) \Rightarrow du = f'(x)dx, \quad x = 1 \rightarrow u = f(1) = 7, \quad x = 3 \rightarrow u = f(3) = 22.$$

The integration by substitution formula gives

$$\int_1^3 f'(x)e^{f(x)}dx = \int_7^{22} e^u du = \left[e^u \right]_{u=7}^{u=22} = e^{22} - e^7.$$

(b) We use the change of variables $u = f(e^x)$,

$$\frac{du}{dx} = f'(e^x) \cdot e^x \Rightarrow du = e^x f'(e^x)dx, \quad x = 0 \rightarrow u = f(1) = 7, \quad x = 1 \rightarrow u = f(e) = 20.$$

The integration by substitution formula gives

$$\int_0^1 e^x f'(e^x)dx = \int_7^{20} du = 13.$$

(c) We use the change of variables $u = f(x)$,

$$\frac{du}{dx} = f'(x) \Rightarrow du = f'(x)dx, \quad x = 0 \rightarrow u = f(0) = 5, \quad x = 1 \rightarrow u = f(1) = 7.$$

The integration by substitution formula gives

$$\int_0^1 f'(x)(f(x))^2dx = \int_5^7 u^2 du = \frac{1}{3} \left[u^3 \right]_{u=5}^{u=7} = \frac{218}{3}.$$

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Problem 2*Compute*

$$\int (2t - 7)^{73} dt$$

Solution.

We use the change of variables $u = 2t - 7$,

$$\frac{du}{dt} = 2 \Rightarrow \frac{1}{2} du = dt.$$

The integration by substitution formula gives

$$\int (2t - 7)^{73} dt = \frac{1}{2} \int u^{73} du = \frac{u^{74}}{148} + C = \frac{(2t - 7)^{74}}{148} + C. \quad \blacksquare$$

Problem 3*Compute*

$$\int t^2 (t^3 - 3)^{10} dt$$

Solution.

We use the change of variables $u = t^3 - 3$,

$$\frac{du}{dt} = 3t^2 \Rightarrow \frac{1}{3} du = t^2 dt.$$

The integration by substitution formula gives

$$\int t^2 (t^3 - 3)^{10} dt = \frac{1}{3} \int u^{10} du = \frac{u^{11}}{33} + C = \frac{(t^3 - 3)^{11}}{33} + C. \quad \blacksquare$$

Problem 4*Compute*

$$\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx$$

Solution.

We use the change of variables $u = \sqrt{x}$,

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow 2du = \frac{1}{\sqrt{x}} dx.$$

The integration by substitution formula gives

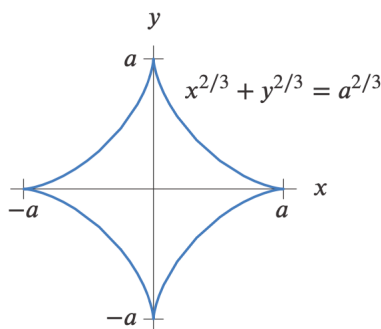
$$\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx = 2 \int \cos(u) du = 2 \sin(u) + C = 2 \sin(\sqrt{x}) + C. \quad \blacksquare$$

Problem 5

Given that

$$\int_0^1 (1 - x^{2/3})^{3/2} dx = \frac{3\pi}{32},$$

compute the area inside the following asteroid



Solution.

By symmetry, the area A of the asteroid is four times the area in its first quadrant,

$$A = 4 \int_0^a (a^{2/3} - x^{2/3})^{3/2} dx = 4 \int_0^a \left(a^{2/3} \left(1 - (x/a)^{2/3} \right) \right)^{3/2} dx = 4a \int_0^a \left(1 - (x/a)^{2/3} \right)^{3/2} dx.$$

To simplify this integral we use the change of variables $u = x/a$,

$$\frac{du}{dx} = \frac{1}{a} \Rightarrow a du = dx.$$

The integration by substitution formula gives

$$A = 4a^2 \int_0^a (1 - u^{2/3})^{3/2} du = \frac{12\pi}{32} a^2 = \frac{3\pi}{8} a^2. \quad \blacksquare$$

Problem 6

Consider a differentiable function F with $f(x) = F'(x)$ defined by the following table of values

x	0	1	2	3	4	5
$f(x)$	2	-5	-6	-1	10	27
$F(x)$	10	8	2	-2	2	20

Determine $\int_0^5 x f'(x) dx$.

Solution.

We can integrate by parts,

$\pm$	D	I
$+$	x	f'
$-\int$	1	f

to get

$$\int_0^5 x f'(x) dx = \left[x f(x) \right]_{x=0}^{x=5} - \int_0^5 f(x) dx = 5f(5) - \int_0^5 f(x) dx.$$

It follows by the First Fundamental Theorem of Calculus that

$$\int_0^5 x f'(x) dx = 5f(5) - (F(5) - F(0)) = 5 \cdot 27 - (20 - 10) = 125. \quad \blacksquare$$

Problem 7

Compute

$$\int t^2 \sin(t) dt$$

Solution.

We can integrate by parts

$\pm$	D	I
+	t^2	$\sin(t)$
−	$2t$	$-\cos(t)$
$+\int$	2	$-\sin(t)$

to get

$$\int t^2 \sin(t) = -\cos(t)t^2 + 2t \sin(t) - \int 2 \sin(t) dt = -\cos(t)t^2 + 2t \sin(t) + 2 \cos(t) + C. \quad \blacksquare$$

Problem 8

Compute

$$\int x^{10} \ln(x) dx$$

Solution.

We can integrate by parts

$\pm$	D	I
+	$\ln(x)$	x^{10}
$-\int$	$\frac{1}{x}$	$\frac{1}{11}x^{11}$

to get

$$\int x^3 \ln(x) dx = \frac{\ln(x)x^{11}}{11} - \frac{1}{11} \int x^{10} dx = \frac{\ln(x)x^{11}}{11} - \frac{x^{11}}{121} + C. \quad \blacksquare$$

Problem 9

Compute

$$\int e^{2x} \cos(x) dx$$

Solution.

We can integrate by parts

$\pm$	D	I
+	e^{2x}	$\cos(x)$
−	$2e^{2x}$	$\sin(x)$
$+\int$	$4e^{2x}$	$-\cos(x)$

to get

$$\int e^{2x} \cos(x) dx = e^{2x} \sin(x) + 2e^{2x} \cos(x) - 4 \int e^{2x} \cos(x) dx.$$

Rearranging reveals that

$$\int e^{2x} \cos(x) dx = \frac{e^{2x}}{5} (\sin(x) + 2\cos(x)) + C. \quad \blacksquare$$

Problem 10For $x \in \mathbb{R}$, the error function is defined by

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

Determine

$$\int \operatorname{erf}(x) dx.$$

Solution.

We can integrate by parts,

$\pm$	D	I
+	$\operatorname{erf}(x)$	1
$-\int$	$\frac{d}{dx} \operatorname{erf}(x)$	x

Since $\frac{d}{dx} \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}$ by the Second Fundamental Theorem of Calculus, we have

$$\int \operatorname{erf}(x) dx = x \operatorname{erf}(x) - \int \frac{2x}{\sqrt{\pi}} e^{-x^2} dx.$$

The second integral can be solved using the substitution $u = -x^2$, $du = -2x dx$ which gives us

$$\int \operatorname{erf}(x) dx = x \operatorname{erf}(x) + \int \frac{1}{\sqrt{\pi}} e^u du = x \operatorname{erf}(x) + \frac{1}{\sqrt{\pi}} \cdot e^{-x^2} + C. \quad \blacksquare$$